

CHL5233: Quarto in-class assignment

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For this in-class assignment, you will work in a group to create your **individual** Quarto document. Please consult your classmates and [Tutorial: Authoring](#) webpage by [Quarto](#). You can also use the Internet including ChatGPT. The TA and I are also available for any questions.

When you are finished, render the document to PDF. Please carefully **proof read** and make edits as needed. **Commit** and **push** the edits to this file and the new PDF file to your **new repository**.

Create a new repository

- Under your GitHub account, create a **new** repository called `armed_conflict`
 - Throughout the course, we will use this repository to **reproduce** and **replicate** the analysis in the paper
- Create **two** folders
 - `data`
 - `reports`
- Save `countries.txt` and `regions.txt` into `data/raw/` subfolder
- Save `quarto_inclass.qmd` into `reports/` folder
- Eventually, your repository will have a very organized structure like this:

Example of data science organization

```
armed_conflict/  
  README.md  
  data/  
    raw/  
    processed/  
  scripts/  
    01_import.R  
    02_clean.R
```

```
03_analysis.R
outputs/
  figures/
  tables/
reports/
  analysis.qmd
```

Edit this file to create a PDF document using Quarto

The R code chunk below includes the R packages required to knit this document. This is often called a **setup** code. We want the code to run, but we don't want the final document to show the code or results. Edit the code chunk so that the code will run but the code and results (including all warnings and messages) are hidden from the final PDF output.

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2     4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr       1.2.2
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become er
```

```
library(knitr)
library(kableExtra)
```

Attaching package: 'kableExtra'

The following object is masked from 'package:dplyr':

```
group_rows
```

For this exercise, we will use the following two data sets.

1. `countries.txt`: A data set that contains a list of countries used in the paper “Implications of armed conflict for maternal and child health: A regression analysis of data from 181 countries for 2000–2019” and their three-digit ISO (International Organization for Standardization) code.

2. `regions.txt`: A data set that contains a list of all major countries in the world and their

- a. ISO code (`country_code`)
- b. region (`region`)
- c. sub-region (`sub_region`)
- d. numerical region code (`region_code`)
- e. numerical sub-region code (`sub_region_code`)

The **final goal** is to display a **bar chart** that depicts how many countries are in each region.

Read in data and create an analytical data set

First, edit the code chunk below that reads in `countries.txt` to also display the number of rows and columns of the data.

```
countries <- read.table("/Users/shanjiejiang/Desktop/CHL5233.nosync/Week 2/Data/Raw_Data/countries.txt")
summary(countries)
```

```
country_name      ISO
Length:186        Length:186
Class :character   Class :character
Mode  :character   Mode  :character
```

```
nrow(countries)
```

```
[1] 186
```

```
count(unique(countries))
```

```
      n
1 186
```

There is 186 Countries in the dataset

Second, read in `regions.txt` and also write a sentence that states how many countries are included in the data set.

```
regions <- read.table("/Users/shanjiejiang/Desktop/CHL5233.nosync/Week 2/Data/Raw_Data/regions.txt")
summary(regions)
```

country_name	country_code	region	sub_region
Length:249	Length:249	Length:249	Length:249
Class :character	Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character	Mode :character

region_code	sub_region_code
Min. : 2.00	Min. : 15.00
1st Qu.: 9.00	1st Qu.: 53.75
Median : 19.00	Median : 154.00
Mean : 65.95	Mean : 179.87
3rd Qu.: 142.00	3rd Qu.: 202.00
Max. : 150.00	Max. : 419.00
NA's : 1	NA's : 1

```
nrow(regions)
```

```
[1] 249
```

```
count(unique(regions))
```

```

n
1 249

```

There is 249 Countries in the dataset

Third, subset the `regions` data set to only include countries from the `countries` data set.

```
regions_subset <- regions[regions$country_code %in% countries$ISO, ]
head(regions_subset)
```

	country_name	country_code	region	sub_region
1	Afghanistan	AFG	Asia	Southern Asia
3	Albania	ALB	Europe	Southern Europe
4	Algeria	DZA	Africa	Northern Africa
6	Andorra	AND	Europe	Southern Europe
7	Angola	AGO	Africa	Sub-Saharan Africa
10	Antigua and Barbuda	ATG	Americas	Latin America and the Caribbean
	region_code	sub_region_code		
1	142	34		

3	150	39
4	2	15
6	150	39
7	2	202
10	19	419

```
write.csv(regions_subset, "/Users/shanjiejiao/Desktop/CHL5233.nosync/Week 2/Data/Processed_Data/re
```

Tabulate countries by region and sub-region

Next, we want to find out how many countries are in a region. Use the `group_by()` and `summarise()` functions in the `dplyr` package to count the number of countries in each region. Then use the `kable()` function in the `knitr` package to create a simple table with the column headers, “Region” and “Count”.

```
region_counts <- regions_subset %>%
  group_by(region) %>%
  summarise(Count = n())

kable(region_counts,
      col.names = c("Region", "Count"))
```

Region	Count
Africa	54
Americas	35
Asia	46
Europe	40
Oceania	11

Create a similar table that shows the number of countries in each **sub-region**.

```
subregion_counts <- regions_subset %>%
  group_by(sub_region) %>%
  summarise(Count = n())

kable(subregion_counts,
      col.names = c("Sub-region", "Count"))
```

Sub-region	Count
Australia and New Zealand	2
Central Asia	5
Eastern Asia	5
Eastern Europe	10
Latin America and the Caribbean	33
Melanesia	4
Micronesia	3
Northern Africa	6
Northern America	2
Northern Europe	10
Polynesia	2
South-eastern Asia	11
Southern Asia	9
Southern Europe	13
Sub-Saharan Africa	48
Western Asia	16
Western Europe	7

Data visualization using ggplot

Here, we will use `geom_bar()` function within `ggplot()` to create a bar chart that displays the **number of countries** in each **sub-region**.

1. Color code each sub-region by **region**.
2. Create bars that run horizontally where sub-region names appear along the y-axis.
3. Order the bars such that the sub-region with the most number countries appear at the top and the sub-region with the least number of countries appear at the bottom.

```
library(ggplot2)

ggplot(regions_subset, aes(x = reorder(sub_region, sub_region, FUN = length),
                             fill = region)) +
  geom_bar() +
  coord_flip() +
  labs(
    x = "Sub-region",
    y = "Number of Countries",
    fill = "Region"
  )
```

